Badrinath Reddy Panyam

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linkedin.com/in/pbnr | Portfolio | Github

EDUCATION

SUNY Stony Brook University, Stony Brook, NY **Jan 2024 – Present**
Master of Science in Data Science ***GPA: 3.89/4.0***
Geethanjali College of Engineering and Technology, Hyderabad, India **Aug 2019 – Jun 2023**
Bachelor of Technology in Computer Science and Engineering ***GPA: 3.53/4.0***
Coursework: Databases, Distributed systems, Operating Systems, Object Oriented Programming, Data Structures, Algorithms, Web Technologies, Computer Vision, Machine Learning, Natural Language Processing, Software Engineering, Computer Networks.

SKILLS

- **Languages:** C, Python, Java, **Go lang**, HTML, CSS, **JavaScript**, **SQL**, XML, PHP, R
- **Development Tools:** Git, Jira, Jenkins, Visual Studio, Postman, MS Office, Eclipse, IntelliJ
- **Technologies:** Bootstrap, jQuery, MVC, SOAP, REST API, **Angular**, Node.js, Typescript, Jest, **Databases**, MySQL, MS SQL, DynamoDB, NoSQL, **Salesforce**, LWC, SLDS, Logging, **Troubleshooting**, Networks, **Distributed Systems**, Hadoop, Spark, Microservices, **NLP**, NumPy, Pandas, PyTorch, TensorFlow, OpenCV, LLM, Flask, **Software Development**, Agile Methodologies, **Low Level Design** (LLD), TDD, **AWS** (Lambda, S3, EC2, SageMaker, Elasticsearch), Docker, Google Cloud Platform, Secure coding, TCP/IP, RDBMS, Linux, CDN, LAMP, BSD, UNIX.

WORK EXPERIENCE

Stony Brook Medicine, Stony Brook University New York, USA
Data Scientist (N3C Enclave) **May 2024 – Present**

- Executed advanced **Data Analysis** exploring the association between **COVID-19** infection and acute kidney injury through robust epidemiological methods, contributing to peer-reviewed clinical research.
- Designed and implemented scalable **ETL** pipelines using **Apache Spark** and **PySpark** for transforming multi-terabyte clinical datasets, reducing processing time by 40%.
- Developed complex **SQL** queries with **Common Table Expressions** (CTEs) and window functions to extract precise patient cohorts from billions of records, improving downstream **Machine Learning** model performance
- Collaborated with cross-functional teams of clinician and data scientists to refine analytical frameworks for healthcare disparity.

ValueLabs Hyderabad, India
Software Engineer I **Jul 2022 – Jan 2024**

- Represented in enterprise-grade projects for clients like **CapitalOne** and **Novatiq**, focusing on scalable, cloud-native solutions, which enhanced system reliability and client satisfaction.
- Redesigned **APIs**, cutting response time by 45% through efficient **REST** and **SOAP** protocols and boosting database performance by 30% through **MS SQL** query optimizations.
- Built responsive and functional UI components using **Angular**, leveraging reusable components, services, and **RxJS** Observables for state management, improving code maintainability by 40%.
- Improved rendering efficiency by 30% by optimizing change detection using **OnPush** strategy and pure pipes, and enhanced **UI** responsiveness with debouncing and throttling techniques using **RxJS**.
- Used Entity **Framework**, **LINQ**, and **ADO.NET** for object-relational mapping, improving data retrieval efficiency by 25% and ensuring seamless interaction with the MS SQL database.
- Implemented backend **microservices** in **Go** for performance-critical features, improving processing latency by 40%.
- Optimized app performance by removing cloud gateway dependencies for direct RESTful communication.
- Used **Postman**, **Swagger** for API testing, ensuring the accuracy, reliability, and consistency of API endpoints.
- Mentored junior developers, conducted code reviews, and led knowledge-sharing sessions on microservices and distributed architectures.

PROJECTS

Distributed Banking System Using Linear PBFT, Distributed Systems, Go **Oct 2024 – Nov 2024**

- Built a scalable Byzantine Fault Tolerant system in **Go**, maintaining safety and liveness under concurrent malicious nodes and failure scenarios, with linear communication and support for 200+ secure transactions across 7-node servers.

Hierarchical Document Classifier, AWS, LLM, Natural Language Processing **Jan 2024 - May 2024**

- Architected a scalable taxonomy enrichment framework using **RoBERTa** entailment detection and **BiLSTM** models, achieving 73%+ classification accuracy.
- Integrated **AWS Lambda**, **S3 OpenSearch**, and **Django** for document retrieval, improving query response times by 30%.

Distributed Tax Management System, Full Stack Application **May 2022 – Jan 2023**

- **Developed** a React + Spring Boot application serving **3,000+ residents**, delivered under **Unnat Bharat Abhiyan** in collaboration with **the Government of Telangana, India**.
- **Automated tax workflows** with high-concurrency processing, ensuring **error-free transactions** and peak-time reliability.
- **Deployed Multimedia Messaging Service alerts**, driving a **30%+ increase** in on-time payments.

Exam Branch OMR Detection Software, Full Stack Application (1st position in College Hackathon) **Jan 2022 – Mar 2022**

- Created a **web application** that automates grading for 15,000+ students, achieving 95% accuracy via **OpenCV**-based **image preprocessing**, saving 400+ faculty hours annually with a full-stack system using **Angular**, and **Spring Boot**.